



MORS

DATA • SPACE • QUANTUM

MORS

Scientific Space Analytics Engine

Autonomous Space Telemetry Auditing,
Anomaly Detection & Verification

تدقيق واكتشاف الشذوذ والتحقق — منصة تحليل الفضاء العلمية

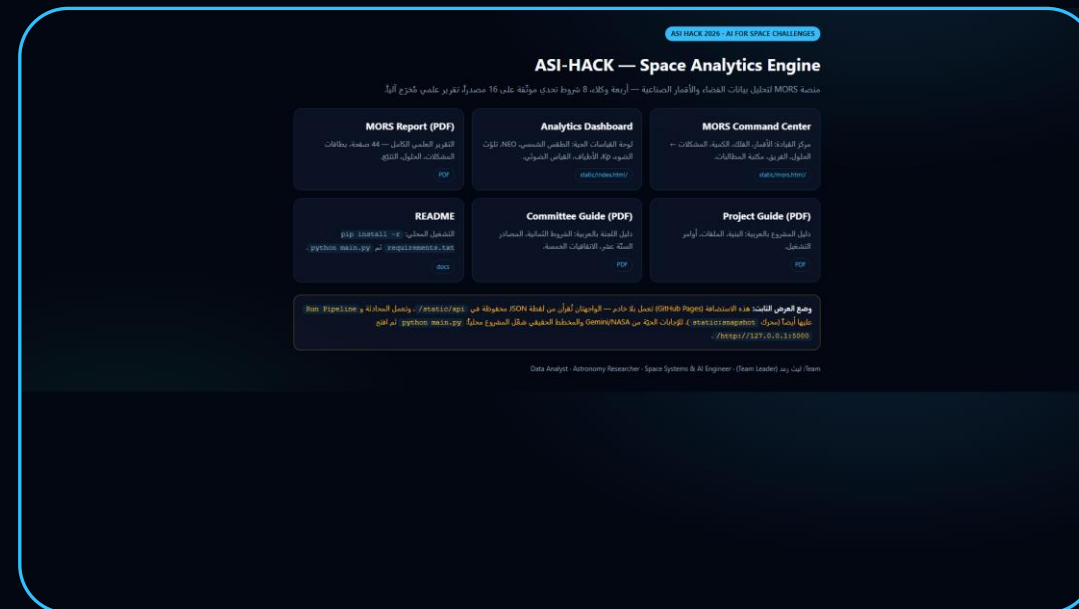
ASI HACK 2026

CERTIFIED 96/100

8/8 CONDITIONS

live deployment • static host • no install

4 agents • 18 modules • 1,711 rows • 225 audit checks • 0 failures
16 curated sources + 12 data sources • report PDF: 44 pages



THE CHALLENGE — VERIFY BEFORE YOU MODEL

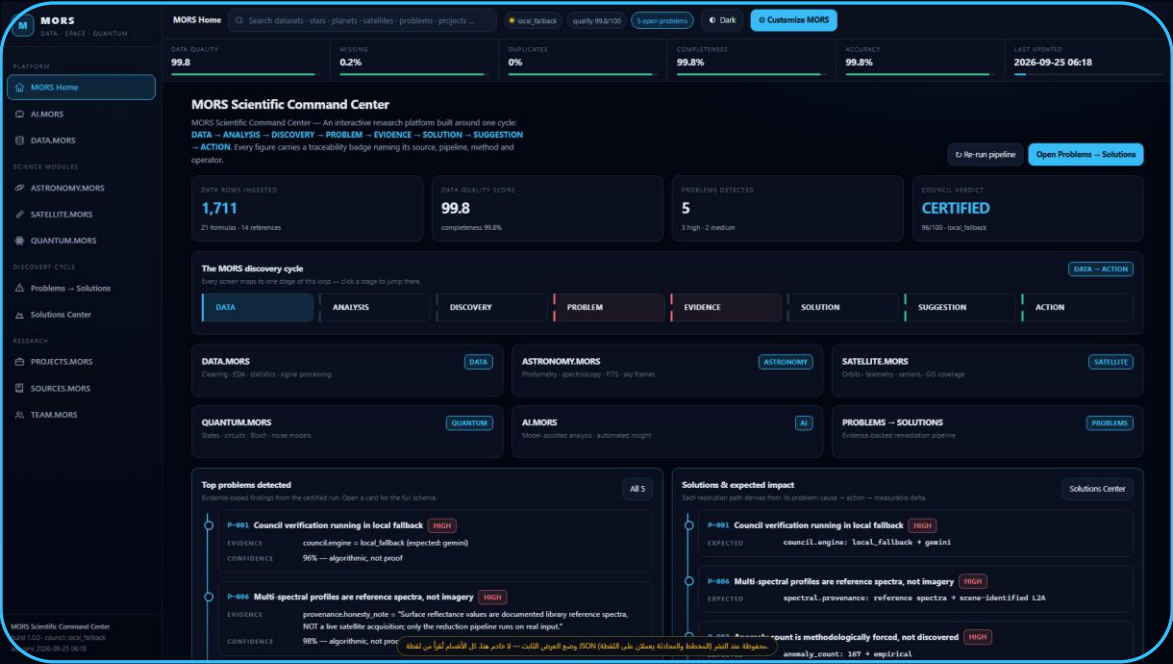
raw volume without verification is noise

Most tools push ML onto telemetry before anyone audits the data itself.

Our own run proved it: the IsolationForest flag rate followed the contamination setting, not real outliers (card P-003).

Data reality after cleaning: 0.192% missing cells · 0 duplicates · 167 QC flags · light-curve S/N 12.5.

Verdict written by the platform: “review-ready, not discovery-grade” — an unfalsifiable model looks scientific while proving nothing.



DATA.MORS data-health · 8,840 cells audited per run





MORS METHODOLOGY — 9-STAGE SCIENTIFIC LIFECYCLE

[prompts/aimors_analyst_prompt.txt](#) · 25 operating rules

1 RAW DATA

ingest + source badge

2 DATA QUALITY

completeness · duplicates · outliers

3 ANALYSIS

EDA · statistics · signals

4 DISCOVERY

patterns worth a question

5 PROBLEM

10-key problem card

6 EVIDENCE

quantified · reproducible

7 SOLUTION

steps + expected delta

8 VERIFICATION

council of five experts

9 RECOMMENDATION

report · trace · publish

TWO HARD RULES

1 · Do not use AI for the sake of AI — if statistics answer the question, use statistics.

2 · Every published value carries source · timestamp · method · operator (traceability badge).

SATELLITE.MORS — FLEET TELEMETRY INSPECTOR

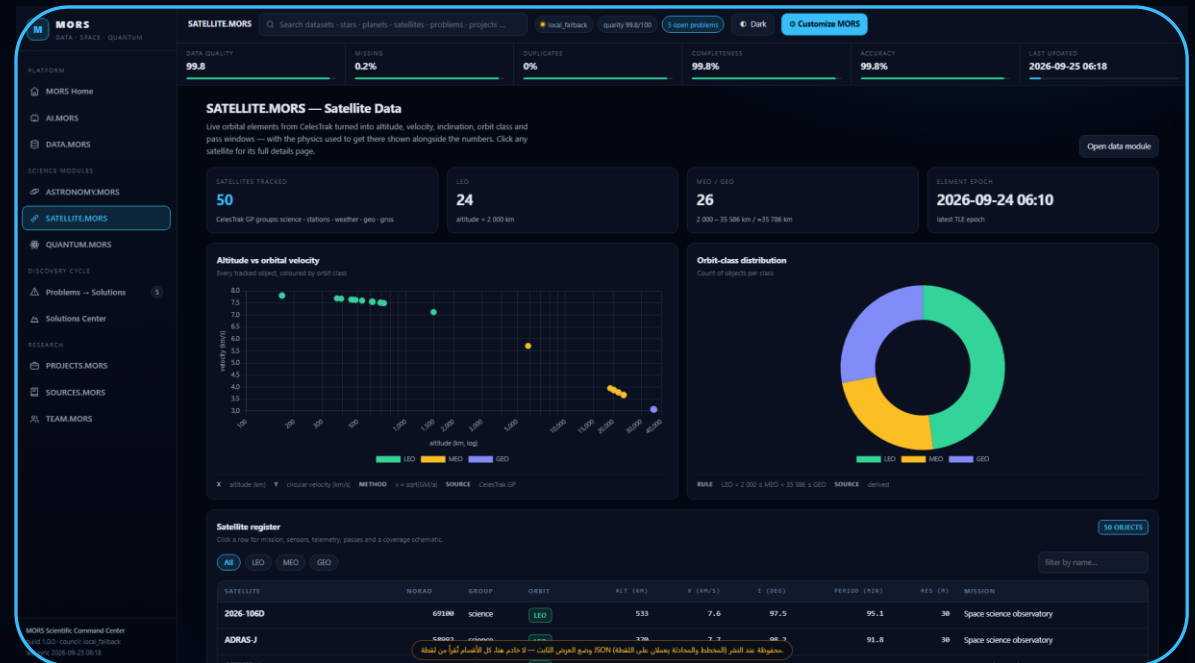
TLE → Keplerian elements → orbit class · live module

50 satellites tracked: altitude 167–36,163 km (LEO → GEO), velocity, inclination, period, perigee/apogee, NORAD ID.

Physics is shown, not hidden: $a = (GM/n^2)^{1/3}$, $v = \sqrt{GM/a}$, $LEO < 2,000 \text{ km} \leq MEO \leq GEO$ belt.

Sensor layer: multispectral imager · 30 m resolution · 290 km swath · telemetry mode · next 5 passes with max elevation.

Spectral bands Blue 490 · Green 560 · Red 665 · NIR 842 · SWIR1 1610 nm → NDVI / NDWI / NDBI + PCA (94.6% variance in 2 components).



SATELLITE.MORS · fleet table + orbit map + band tabs

SATELLITES

50

ALT RANGE

167–36,163
km

RESOLUTION

30 m

SWATH

290 km

PROBLEMS → SOLUTIONS ENGINE

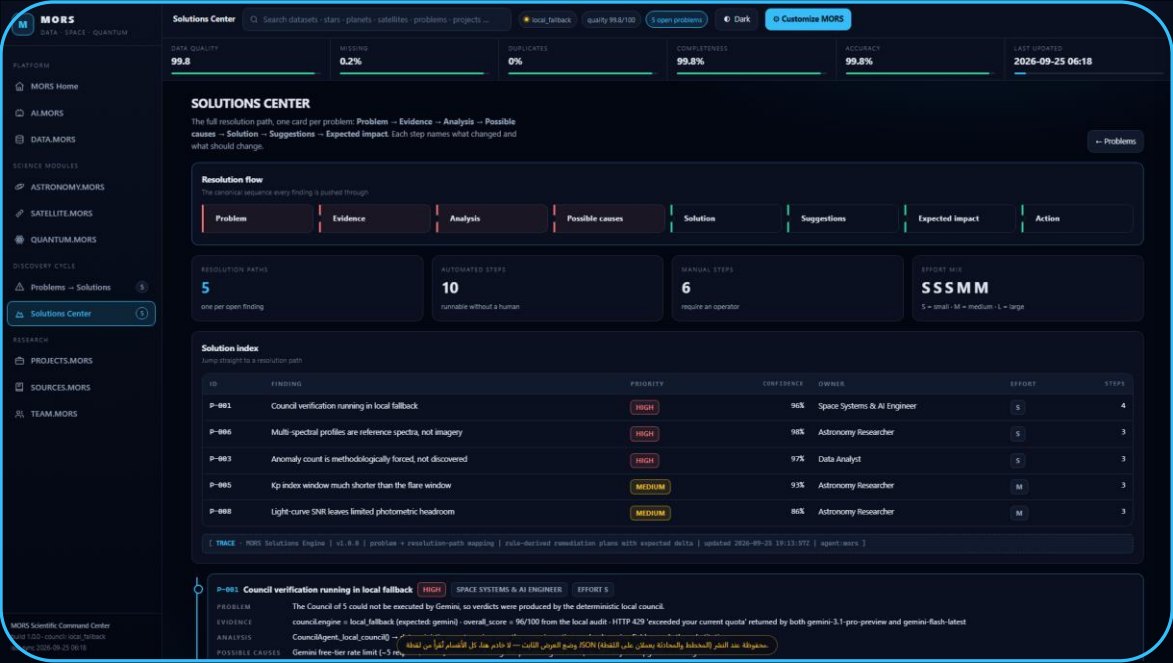
evidence · root cause · fix · measurable delta

Every anomaly is a 10-key card: symptom · root cause · fix · evidence · severity · category · component · priority code.

Priority is arithmetic, not opinion: $0.40 \cdot \text{evidence} + 0.35 \cdot \text{impact} + 0.25 \cdot \text{confidence}$, printed on the card.

Live card P-001: council runs local_fallback under Gemini 429 → expected delta council.engine: local_fallback → gemini.

Live card P-006: spectral indices are reference-labelled → delta spectral.provenance: reference spectra → scene-identified L2A.



SOLUTIONS.MORS · path, steps, effort, expected delta



TRUSTWORTHY AI — CONFIDENCE TAXONOMY

what the platform actually prints on every answer

STANDARD AI PIPELINE

- Model runs first, data quality never inspected.
- Single number published, no error bounds.
- No source, no timestamp, no operator recorded.
- Failure mode looks like a confident answer.
- Cannot be re-run or falsified by a judge.

MORS XAI ENGINE

- Data audit first — ML only when statistics are insufficient.
- Confidence level + priority formula published per card.
- Every value carries source · timestamp · method · operator.
- Engine honesty: local_fallback is labelled, never disguised.
- Council of five experts cross-checks and corrects claims.

5-TIER PROTOCOL ON EVERY AI ANSWER

[Observed Data]

[Calculated Metric]

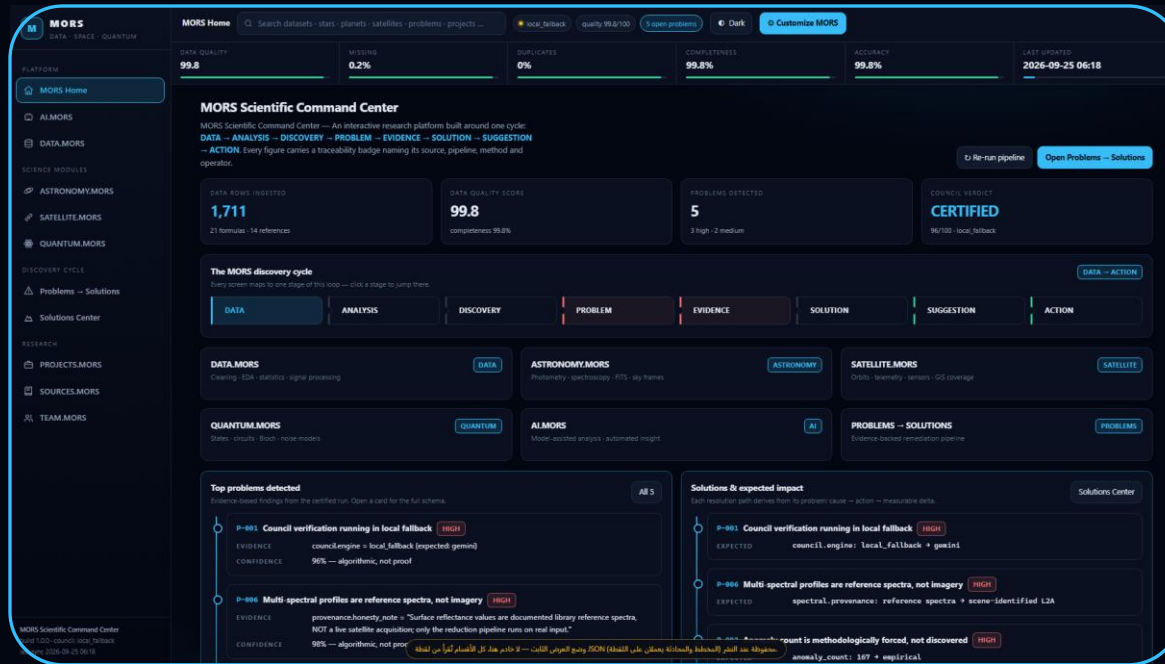
[AI Interpretation]

[Hypothesis]

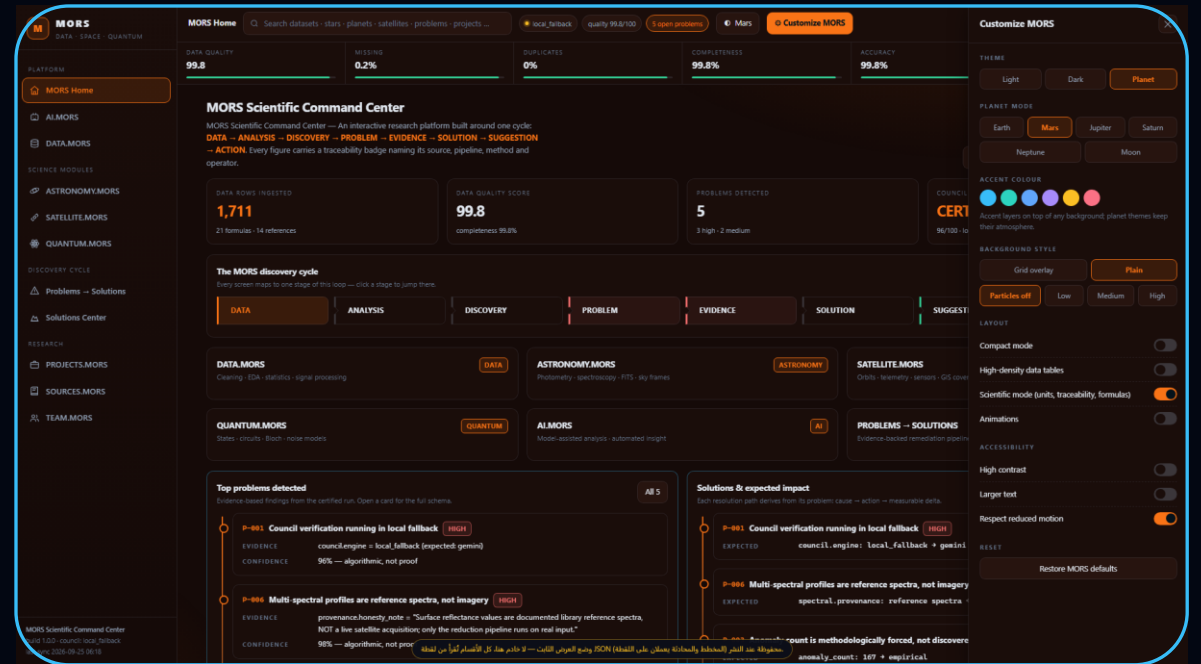
[Suggested Action]

HIGH-DENSITY SCIENTIFIC UI / UX

dark glassmorphism · Orbitron + IBM Plex Mono + Cairo



EARTH CYAN · dark glass panels · lifecycle ribbon



MARS RUST · planet theme engine · one click

Planet themes: Earth · Mars · Jupiter · Saturn · Neptune · Moon,
plus light/dark, accent layers, grid and particle toggles.

Accordion sidebar: DATA · ASTRONOMY · SATELLITE · QUANTUM · AI ·
PROBLEMS→SOLUTIONS, with global search and instant switching.

ENGINEERING CAPABILITIES — TEAM MATRIX

4 members · published roles · module ownership

TEAM LEADER

ليث رعد · Layth Ra'ad

System Architecture · Pipeline Orchestration · Scientific Computing · Technical Review · Scientific Visualization

owns: MORS architecture · HOME · PROJECTS · TEAM

DATA ANALYST

role seat

Data Cleaning · EDA · Statistical Analysis · Time Series · Anomaly Detection · SQL · Feature Engineering

owns: DATA.MORS · PROBLEMS.MORS · SOLUTIONS.MORS

ASTRONOMY RESEARCHER

role seat

Astropy · FITS Analysis · Photometry · Spectroscopy · Light-Curve Analysis · Sky Coordinates

owns: ASTRONOMY.MORS · SOURCES.MORS

SPACE SYSTEMS & AI ENGINEER

role seat

Satellite & Orbit Analysis · AI Model Integration · Quantum Computing · Remote Sensing · Scientific Computing

owns: AI.MORS · SATELLITE.MORS · QUANTUM.MORS

MEASURABLE PLATFORM IMPACT

every number below is printed by the live platform

225

AUTOMATED CHECKS

0 failures · ~1 min run

99.81%

DATA QUALITY

8,840 cells · 1,711 rows

100%

TRACEABILITY

source · time · method · operator

96/100

COUNCIL VERDICT

CERTIFIED · 5 experts

SUPPORTING EVIDENCE

- 21 cited formulas · 14 references
- 8 live endpoints (NASA · NOAA · CelesTrak · GIBS)
- 18 UI modules · 19 registered endpoints
- 5 problem cards · 10-key schema each
- 16 curated sources + 12 data sources
- 8/8 challenge conditions (7 core + 1 extension)
- 6 published master prompts
- 44-page scientific report, auto-generated

THE FUTURE OF SPACE DATA ANALYTICS

build · learn · innovate

Try the live engine — no install, no account:

<https://laythraad.github.io/ASI-HACK-Space-Analytics-Engine/>

Next: deeper NASA / ESA live integration,
real Sentinel-2 L2A scene ingestion for the spectral module,
and scaling the quantum block beyond the classical statevector
(labelled honestly today as simulation, never as QPU output).

CERTIFIED 96/100 · 225 checks · 0 failures · 8/8 conditions
Open the link above and click through every module yourself.



SCAN → LIVE DEMO

Team

ليث رعد (Team Leader)

Data Analyst · Astronomy Researcher ·
Space Systems & AI Engineer